

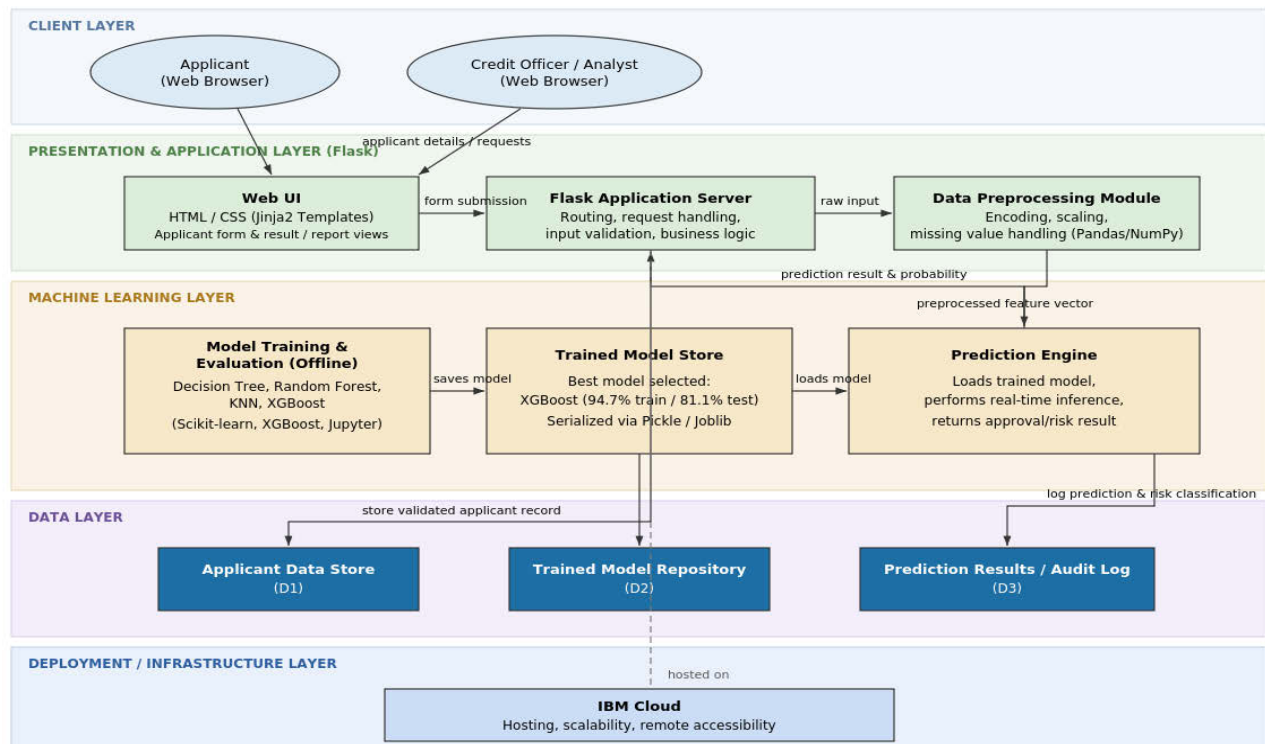
## Solution Architecture

|                      |                 |
|----------------------|-----------------|
| <b>Date</b>          | 28-06-2026      |
| <b>Team ID</b>       | SWTID-2026-3845 |
| <b>Project Name</b>  | SMART LENDER    |
| <b>Maximum Marks</b> | 5 Marks         |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

#### Smart Lender - Solution Architecture



## Component Description Table

| Component Name                        | Description / Role in Architecture                                   | Technologies Used                          |
|---------------------------------------|----------------------------------------------------------------------|--------------------------------------------|
| <i>Web UI</i>                         | <i>Screen where applicants enter details and staff view results.</i> | <i>HTML, CSS, Jinja2 (Flask templates)</i> |
| <i>Flask Application Server</i>       | <i>manages flow between UI and ML model.</i>                         | <i>Python, Flask, Pandas, NumPy</i>        |
| <i>Prediction Engine</i>              | <i>Predicts loan approval or rejection using the trained model.</i>  | <i>Python, XGBoost, Scikit-learn</i>       |
| <i>Data Store &amp; Cloud Hosting</i> | <i>Stores records, model, and logs; hosts the app online.</i>        | <i>Database (SQLite), IBM Cloud</i>        |